AI-Driven Multimodal Classification of Brain Tumours Using MRI and Gene Expression Data

Complete Technical Report

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1. Introduction and Motivation

Brain tumours represent one of the most clinically challenging classes of neoplasms, with diagnosis traditionally relying on cranial neuroimaging (MRI) and postoperative histopathological analysis. The advent of high-throughput transcriptomics has established that molecular signatures provide essential diagnostic and prognostic stratification that supersedes macroscopic morphology alone.

This project implements an **AI-driven multimodal diagnostic framework** structured into three clearly distinguished operational pathways:

- **MRI-Only Pathway:** Structural classification across 4 classes: Glioma, Meningioma, Pituitary Tumour, and No Tumour. An approximately 90% accuracy level is used as the project evaluation target/reference for the primary DenseNet121 architecture, with spatial interpretability supplied via Grad-CAM.
- **Gene-Expression-Only Pathway:** Transcriptomic classification across 4 tumour classes: LGG (Lower-Grade Glioma), GBM (Glioblastoma Multiforme), Meningioma, and Pituitary Tumour. Evaluated on an integrated multi-cohort dataset (793 samples across 100 features) using Random Forest, XGBoost, and MLP classifiers, with feature importance explained via TreeSHAP.
- **True Multimodal / Radiogenomic Pathway:** Synergistic late fusion combining cranial MRI and transcriptomic profiling from the **SAME verified patient**, enforced through strict barcode validation (TCGA-XX-XXXX) and restricted to diffuse gliomas (GBM vs LGG).

The system provides three modular diagnostic pathways deployed through an interactive Streamlit clinical web platform with automated medical PDF report generation.

2. Literature Review and Background

2.1 Brain Tumour Classification via MRI

Convolutional Neural Networks (CNNs) have achieved strong results in brain tumour classification from MRI scans [1-6]. Transfer learning from ImageNet-pretrained backbones (DenseNet121, EfficientNetB0, ResNet50) enables effective training even with moderate dataset sizes by leveraging hierarchical visual feature representations. Foundational benchmarks established by Cheng et al. [1] on contrast-enhanced T1-weighted MRI (CE-MRI) demonstrated the feasibility of separating glioma, meningioma, and pituitary tumours. Subsequent investigations by Deepak & Ameer [2], Swati et al. [3], Badža & Barjaktarovi■ [4], and Noreen et al. [5] further established that deep neural representations substantially surpass handcrafted morphological descriptors.

2.2 Molecular Subtyping via Gene Expression

The Cancer Genome Atlas (TCGA) Research Network established through landmark studies, notably Ceccarelli et al. [7], that molecular profiles (IDH mutation status, 1p/19q codeletion, and transcriptomic signatures) capture biologically discrete subsets of diffuse glioma that correlate more robustly with patient survival than classical histological grade alone. In parallel, public repositories such as GEO provide gene expression series for non-glioma cranial tumours (GSE43290 for meningioma, GSE93825 for pituitary adenoma), enabling multi-cohort integration for four-class transcriptomic classification.

2.3 Radiogenomics and Multimodal Fusion

Radiogenomics bridges macro-scale radiological imaging phenotypes with micro-scale genomic and molecular characteristics. Standardised reference cohorts established by Bakas et al. [9] through TCIA and TCGA demonstrated that quantitative imaging features correlate with underlying genetic alterations, including MGMT promoter methylation (Korfiatis et al. [8]). Concurrently, multimodal fusion frameworks (Mobadersany et al. [10], Cheerla & Gevaert [11], Chen et al. [12]) have demonstrated the prognostic benefit of combining visual phenotypes with transcriptomic data.

3. Comparison with Existing Studies

To rigorously position this project within the state of the art, we surveyed peer-reviewed literature across leading biomedical, computer science, and medical imaging venues (IEEE, Elsevier, Springer, Nature Portfolio, Cell Press, PNAS, and PubMed/PMC). Studies were selected to cover seven foundational methodological dimensions: (1) MRI-only deep learning, (2) multi-class MRI classification, (3) gene-expression / molecular classification, (4) TCGA–TCIA radiogenomics, (5) multimodal MRI + molecular analysis, (6) explainable AI for neuroimaging, and (7) explainable AI for molecular genomics.

The table below provides a detailed, objective comparison across all thirteen selected peer-reviewed studies and our current project implementation. In accordance with strict scientific reporting standards, no rankings or unverified claims are made. All values reflect factual parameters from published literature and verified repository artifacts.

Study	Year	Dataset	Modality	Task	Classes / Cohort	Model / Method	Fusion	XAI	Reported Result	Limitation / Gap	Relation to Our Study
Cheng et al. [1]	2015	Nanfang/Tianjin (3,064 slices)	MRI (T1-CE)	Classification	Glioma, Mening., Pituitary (3)	Region augment. + BoVW + SVM	None	None	91.28% Acc	Handcrafted features; 2D slice; no genomics	Baseline 3-class CE-MRI benchmark
Deepak & Ameer [2]	2019	Cheng (Figshare) (3,064 slices)	MRI (T1-CE)	Classification	Glioma, Mening., Pituitary (3)	GoogLeNet + SVM	None	None	98.0% Acc (5-fold)	MRI-only; no no-tumour control; black-box	Demonstrates transfer learning on brain MRI
Swati et al. [3]	2019	Cheng (Figshare) (3,064 slices)	MRI (T1-CE)	Classification	Glioma, Mening., Pituitary (3)	Fine-tuned VGG19	None	None	94.82% Acc	Heavy backbone; slice-level; no XAI	Transfer learning comparison baseline
Badža & Barjaktarovi■ [4]	2020	Cheng (3,064 slices)	MRI (T1-CE)	Classification	Glioma, Mening., Pituitary (3)	Custom 22-layer CNN	None	None	96.56% Acc	Trained from scratch; sensitive to augment.	Custom CNN benchmark on CE-MRI
Noreen et al. [5]	2020	Figshare (3,064 slices)	MRI (T1-CE)	Classification	Glioma, Mening., Pituitary (3)	DenseNet201 + InceptionV3 concat.	Intermediate (imaging)	None	99.34% Acc	Concatenated feature bloat; no genomics	Validates DenseNet feature representations
Díaz-Pernas et al. [6]	2021	Cheng (3,064 slices)	MRI (T1-CE)	Classif. & Segment.	Glioma, Mening., Pituitary (3)	Multiscale CNN + spatial pyramid	None	Segmentation mask	97.3% Acc	Requires manual segmentation masks	Localisation comparison (masks vs Grad-CAM)
Ceccarelli et al. [7]	2016	TCGA Pan-Glioma (1,122 patients)	Multiplatform Genomics	Subtyping & Prognosis	IDH-mut (codel/noncodel) vs IDH-wt	Consensus clustering + Random Forest	Multi-omics (genomic)	Gene heatmaps	5 molecular subtypes defined	Tissue biopsy needed; no MRI correlation	Ground truth molecular basis for our gene model
Korfiatis et al. [8]	2017	TCIA Glioblastoma (155 patients)	MRI (T2-weighted)	MGMT Radiogenomics	MGMT Methylated vs Unmethylated	Deep residual CNN (ResNet-style)	Unpaired radiogenomics	Not reported	94.9% Acc	Single-centre T2; single gene target	Shows MRI-to-gene predictive feasibility
Bakas et al. [9]	2017	TCGA-TCIA (285 subjects)	Multi-sequence MRI + Genomics	Radiomic Profiling	GBM (102) vs LGG (183)	704 radiomic features + expert labels	Correlative mapping	Radiomic stats	Benchmarked TCGA-TCIA radiogenomics	Extensive manual pre-processing required	Standard for our verified patient pairing
Mobadersany et al. [10]	2018	TCGA Glioma (769 patients)	Histology WSI + Genomics	Survival Prediction	Diffuse glioma risk tiers	GSCNN (CNN + Cox proportional hazards)	Intermediate concat.	Histology heatmaps	C-index 0.754–0.841	Invasive histology only; no cranial MRI	Multimodal vision + genomic integration
Cheerla & Gevaert [11]	2019	TCGA Pan-Cancer (11,160 pts, 20 types)	Clinical + mRNA + miRNA + WSI	Pan-cancer Prognosis	Pan-cancer risk groups	Highway Nets + CNN + Resilient Dropout	Intermediate latent fusion	Modality dropout	C-index 0.784	No MRI neuroimaging; complex multi-omics	Highlights missing-modality handling
Chen et al. [12]	2022	TCGA GBMLGG & KIRC (1,014 pts)	Histology WSI + Genomics	Prognosis & Grade Classif.	Low-grade vs High-grade glioma	Pathomic Fusion (Bilinear Tensor + Attention)	Bilinear tensor fusion	Attention heatmaps	C-index 0.826	Invasive histology-based; no clinical UI	Tensor fusion benchmark vs late fusion
Zeineldin et al. [13]	2022	BraTS Benchmark	Multi-sequence MRI	Tumour Segmentation	Whole, core, enhancing tumour	Deep U-Net + Grad-CAM variants	None	Grad-CAM, Guided Grad-CAM	High visual agreement	Imaging only; no genomic profiling	Confirms Grad-CAM reliability in brain MRI
OUR STUDY / CURRENT PROJECT	**2026**	**BDNeuro-MRI (7,023) + TCGA/TCIA (793 genes, 682 mapped pairs) + GEO**	**Multimodal (MRI + RNA-seq Gene Expression)**	**4-Class MRI + 4-Class Gene + Verified Multimodal Fusion**	**MRI: Glioma, Mening., Pituit., NoTum; Gene: LGG, GBM, Mening., Pituit.**	**DenseNet121 (MRI) + RF/XGBoost (100 Genes)**	**Conditional Late Fusion (0.45 MRI + 0.55 Gene)**	**Dual XAI: Grad-CAM + TreeSHAP**	**Target ~90% MRI ref. (77.50% test bench, final test not verified); Gene: 100% test acc; Fusion: 100% on verified pairs**	**2D axial slices; BDNeuro not patient-linked to TCGA; fusion restricted to verified gliomas**	**Integrates 4-class MRI, 4-class genomics, clinical verification gate, dual XAI, Streamlit UI**

3.3 Methodological Analysis Across Key Dimensions

1. Dataset Dimensions & Cohort Scale: Benchmark MRI studies [1-6] predominantly evaluated the Cheng dataset (3,064 slices, 233 patients) without a 'no tumour' control cohort. The current project evaluates 7,023 images across 4 balanced classes (BDNeuro) for general structural triage, paired with 43 verified patient-matched TCGA/TCIA cases. For genomics, our project implements a 4-class transcriptomic dataset (793 samples across 100 features) combining TCGA-LGG (530), TCGA-GBM (172), GEO GSE43290 Meningioma (51), and GEO GSE93825 Pituitary (40).

2. Imaging Modalities & Preprocessing: Prior works used single-sequence T1-CE [1-6] or multiparametric BraTS volumes [9, 13]. Our current implementation processes 2D axial MRI slices normalized with ImageNet statistics and CLAHE.

3. Molecular & Genomic Signatures: While prior radiogenomics focused on single targets like MGMT methylation [8], our study performs multi-cohort feature selection across transcriptomic profiles, isolating a 100-gene panel.

4. Task Formulation: Prior work bifurcated into structural classification [1-6] or survival prediction [10-12]. Our work integrates both: primary structural triage (4 classes) and transcriptomic classification (4 classes).

5. Model Architectures & Parameter Efficiency: Unlike heavy VGG backbones (143M params) [3] or concatenated networks [5], our framework utilizes parameter-efficient DenseNet121 (7.3M params) and fast tree ensembles (RF/XGBoost).

6. Multimodal Fusion Strategy: Rather than raw concatenation or complex bilinear tensor products [12], our project implements modular late probability fusion (0.45 MRI + 0.55 Gene), enabling standalone unimodal fallback if genomics is absent.

7. Patient-Level Pairing & Clinical Verification Gate: In many prior studies, cohort-level associations were assumed. In our project, **TCGA–TCIA patient-level verification is strictly required before multimodal fusion is performed.** Barcode validation (TCGA-XX-XXXX) and histological glioma concordance prevent cross-patient or cross-category data contamination.

8. Explainability & Clinical Transparency: Where prior works provided either Grad-CAM [13] or genomic SHAP independently, our project unifies dual XAI: Grad-CAM on DenseNet121 layer *conv5_block16_2_conv* alongside TreeSHAP on the 100-gene signature.

3.4 Identified Research Gaps

Existing Approach	Limitation Identified	How Our Project Addresses It	Current Limitation
MRI-Only Deep Learning [1-6]	Lacks molecular resolution; cannot separate IDH-mutant LGG from IDH-wt GBM	Integrates RNA-seq gene expression profiling via Random Forest and XGBoost	Genomic testing requires tissue sample; falls back to MRI triage if absent
Gene-Only Classifiers [7]	Invasive biopsy required; lacks macroscopic cranial anatomical context	Pairs genomics with non-invasive cranial MRI to corroborate structural phenotypes	Biopsy still necessary to obtain transcriptomic profile in practice
Unpaired Multimodal Fusion	Risks clinically invalid fusion across mismatched patients or tumour types	Strict TCGA-TCIA barcode and glioma concordance verification gate	Restricts multimodal late fusion to verified paired glioma cases (682 mapped pairs)
Black-Box Multimodal Models [10-12]	End-to-end or tensor fusion obscures individual modality contributions	Deploys dual XAI: Grad-CAM (spatial) and TreeSHAP (biomarker attributions)	Attributions reflect correlation rather than direct mechanistic oncogenic causality
Image-Level Evaluation	Slice-level splits risk data leakage across patient scans	Enforces patient-level verification on TCGA-TCIA pairs explicitly	Public BDNeuro MRI benchmark is slice-level; patient metadata is TCGA-only

3.5 Positioning of the Current Study

Our study synthesises multiple validated components into an integrated clinical decision-support framework. We explicitly distinguish implemented features from planned future extensions:

- **Implemented:** 4-class MRI classification (DenseNet121 on 7,023 images; target ~90% ref., 77.50% test bench, final test not verified); 4-class gene expression subtyping (LGG, GBM, Meningioma, Pituitary Tumour on 793 samples); TCGA-TCIA barcode verification gate; conditional late fusion (0.45/0.55); dual Grad-CAM & TreeSHAP XAI; Streamlit clinical web app with automated PDF reporting.
- **Planned / Future Work:** 3D volumetric DICOM/NIfTI segmentation; co-registration of multi-sequence MRI (T1, T1-Gd, T2, FLAIR); multi-omics integration (DNA methylation, IDH1/2 mutations); prospective multi-centre trials.

3.6 Critical Distinction Between Data Pathways

Important Scientific Clarifications:

- **Four-Class MRI Pathway:** Uses the BDNeuro-MRI benchmark (Glioma, Meningioma, Pituitary Tumour, No Tumour) for structural screening. *BDNeuro images are NOT TCGA–TCIA scans and cannot be directly paired with TCGA gene-expression samples.*
- **Radiogenomic Pathway:** Uses matched TCGA RNA-seq and TCIA MRI scans for diffuse glioma patients with verified barcodes (TCGA-XX-XXXX). *Not all four MRI classes have gene data; radiogenomic fusion is restricted to verified gliomas.*

• **Legacy Demo Deprecation:** The early 5-pair index-aligned demonstration is deprecated and excluded from clinical reporting as it does not represent genuine biological multimodal evidence.

3.7 Visual Summary of Research Positioning

3.8 Synthesis and Summary of Comparison

1. Previous MRI-only studies demonstrated the power of deep learning for structural triage, but cannot provide molecular insight.
2. Previous genomic studies proved transcriptomic signatures define distinct glioma subclasses, but require surgical tissue biopsy.
3. TCGA–TCIA radiogenomics established that macroscopic imaging features correlate with underlying molecular alterations.
4. Multimodal studies demonstrated prognostic advantages from combining vision and genomics, but often relied on histology rather than MRI.
5. Explainability methods (Grad-CAM and SHAP) have been applied independently, but rarely combined across modalities in a unified tool.
6. Our project integrates 4-class MRI, 4-class transcriptomic subtyping, patient barcode verification, late fusion, dual XAI, and a Streamlit UI.
7. Key project limitations include 2D slice processing, unlinked BDNeuro-TCGA pairing, and unverified final test accuracy at the target ~90% MRI level.

4. System Architecture Overview

The system architecture implements three clearly separated diagnostic pathways: (1) MRI-Only Pathway (DenseNet121, 4 classes), (2) Gene-Only Pathway (RF/XGBoost, 4 classes), and (3) Multimodal Radiogenomics Pathway (verified TCGA barcode required, GBM vs LGG).

Key Source Modules

Module	File	Purpose
MRI Model	src/mri_model.py	CNN architectures, training, evaluation
Gene Model	src/gene_model.py	4-class RF/XGBoost/MLP training and inference
Multimodal Fusion	src/multimodal_model.py	Late fusion classifier, patient verification
Prediction Router	src/predictor.py	Unified BrainTumourPredictor API routing 3 modes
XAI Utilities	src/xai_utils.py	Grad-CAM and SHAP computations
Feature Selection	src/feature_selection.py	Multi-method gene feature selection
TCGA Mapping	src/tcga_tcia_mapping.py	Patient barcode verification (682 pairs)
Configuration	src/config.py	Cross-platform path management

5. Datasets

5.1 MRI Neuroimaging Dataset (BDNeuro)

Split	Glioma	Meningioma	Pituitary Tumour	No Tumour	Total
Training	1,400	1,400	1,400	1,400	5,600
Testing	400	400	400	400	1,600
Total	1,800	1,800	1,800	1,800	7,200

- Resolution: Standardized to 224×224 RGB
- Corrupted images: 0 (all verified clean)
- Class balance: Balanced across all 4 classes
- **Scientific Clarification:** BDNeuro-MRI is anonymous slice-level data, does NOT contain TCGA patient barcodes, and cannot be paired with TCGA gene-expression samples.

5.2 Gene Expression Datasets (Four-Class Cohort)

Dataset	Cohort Source	Samples	Tumour Class	Platform
TCGA-LGG	The Cancer Genome Atlas	530	LGG (Lower-Grade Glioma)	RNA-seq (Illumina HiSeq)
TCGA-GBM	The Cancer Genome Atlas	172	GBM (Glioblastoma Multiforme)	RNA-seq (Illumina HiSeq)
GSE43290	Gene Expression Omnibus	51	Meningioma	Microarray (Affymetrix)
GSE93825	Gene Expression Omnibus	40	Pituitary Tumour	Microarray (Affymetrix)
Total	Multi-Cohort Integration	793	4 Tumour Classes	Harmonised 100-Gene Panel

6. Data Preprocessing Pipeline

6.1 MRI Preprocessing

Step	Method	Details
Resize	Bilinear interpolation	224×224 pixels

Step	Method	Details
Format	RGB conversion	3-channel uint8
Normalisation	ImageNet preprocessing	DenseNet-specific mean/std
Quality Control	Motion blur detection	Corrupt images rejected
CLAHE	Contrast enhancement	Applied during training to enhance lesion edges

6.2 Gene Expression Preprocessing

Step	Method	Details
Normalisation	Z-score (StandardScaler)	Per-cohort, zero mean, unit variance
Cross-platform alignment	Feature union	Intersection of gene symbols
Missing values	Zero imputation	For absent features
Label encoding	LabelEncoder (4 classes)	0=GBM, 1=LGG, 2=Meningioma, 3=Pituitary

7. Feature Selection Methodology

Method	Type	Implementation
Variance Threshold	Filter	Remove near-zero variance genes
ANOVA F-Test	Filter	sklearn.feature_selection.f_classif
Mutual Information	Filter	sklearn.feature_selection.mutual_info_classif
Recursive Feature Elimination	Wrapper	RFE with LogisticRegression estimator
Harmonized Selection	Ensemble	Top 100 features saved in gene_feature_names.json

- Top-100 feature set: **100 genes** used for 4-class model training
- Artefact persisted: models/gene_feature_names.json

8. Model Architectures

8.1 MRI Classification Models

Model	Backbone	Parameters	Input	Output	Saved As
DenseNet121 (Production)	ImageNet	7,301,956	224×224×3	4-class softmax	best_mri_model.keras
EfficientNetB0	ImageNet	4,379,559	224×224×3	4-class softmax	efficientnetb0.keras
ResNet50	ImageNet	24,114,308	224×224×3	4-class softmax	resnet50.keras
Custom CNN	None	1,970,628	224×224×3	4-class softmax	custom_cnn.keras

8.2 Gene Expression Models

Model	Algorithm	Key Parameters	Input	Output Classes
Random Forest	sklearn RF	500 trees, balanced weights	100 genes	4 classes (LGG, GBM, Mening., Pituit.)
XGBoost	XGBClassifier	400 trees, depth=6, lr=0.05	100 genes	4 classes (LGG, GBM, Mening., Pituit.)
MLP	sklearn MLP	256→128→64, ReLU, Adam	100 genes (scaled)	4 classes (LGG, GBM, Mening., Pituit.)

8.3 Multimodal Late-Fusion Classifier

- Architecture: **LateFusionClassifier** — calibrated probability soft-voting
- MRI Component: DenseNet121 4-class predictions
- Gene Component: RF/XGBoost ensemble 4-class predictions
- Fusion Formula: P_fusion = 0.45 * P_MRI + 0.55 * P_Gene (glioma-subspace only)
- Eligibility: Verified TCGA barcode + MRI glioma prediction

9. Training Protocol

9.1 MRI CNN Training

- Framework: TensorFlow 2.19.0 / Keras 3.15.1
- Platform: Google Colab (GPU-accelerated)
- Data Split: 5,600 Training / 1,600 Testing
- Optimiser: Adam | Loss: Categorical Cross-Entropy
- Callbacks: EarlyStopping, ReduceLROnPlateau

9.2 Gene Model Training

- Framework: scikit-learn, XGBoost
- Dataset: 793 samples across 100 features (four_class_genes.csv)
- Cross-Validation: 5-fold Stratified K-Fold
- Scaling: StandardScaler fitted per-cohort and merged
- Class Weighting: Balanced (sklearn compute_class_weight)

10. Experimental Results

10.1 MRI Model Comparison (Multi-Model Benchmark)

Model Architecture	Test Accuracy	Macro Precision	Macro Recall	Macro F1	Macro ROC-AUC	Status / Notes
DenseNet121 (Production)	0.7750	0.7921	0.7750	0.7630	0.9429	Target ~90% ref.; initial bench: 77.50%; final test not verified
EfficientNetB0	0.8325	0.8331	0.8325	0.8271	0.9505	Benchmark champion in Phase 5 report (48.5s train)
ResNet50	0.8025	0.8134	0.8025	0.7930	0.9427	Deep residual baseline (24.1M parameters)
Custom CNN	0.2500	0.0625	0.2500	0.1000	0.5442	Failed to learn without transfer learning

Scientific Accuracy Note: An approximately 90% accuracy level is used as the project evaluation target/reference for DenseNet121. In the initial multi-model benchmark evaluation on the held-out test cohort (400 scans), DenseNet121 achieved 77.50% test accuracy (ROC-AUC 0.9429) across the 4 classes, while intermediate training validation reached 90.00%–91.67%. The final clinical test accuracy at the target ~90% level is marked as *not independently verified*.

10.2 Gene Expression Model Results (Four-Class Benchmark)

Model	Task	Accuracy	Macro F1	ROC-AUC	MCC	Cohen's Kappa
Random Forest	4-Class (LGG/GBM/Men/Pit)	1.0000	1.0000	1.0000	1.0000	1.0000
XGBoost	4-Class (LGG/GBM/Men/Pit)	1.0000	1.0000	1.0000	1.0000	1.0000
MLP	4-Class (LGG/GBM/Men/Pit)	0.9645	0.9645	0.9935	0.9047	0.9002

10.3 Multimodal Fusion Results

Model	Task	Accuracy	F1 (Macro)	ROC-AUC	Eligibility Gating
Late-Fusion Ensemble	Verified Glioma Radiogenomics	1.0000	1.0000	1.0000	Enforced: TCGA Barcode + Glioma match

10.4 Diagnostic Mode Comparison Summary

Mode	Classes Handled	Champion Model	Measured Performance	Target / Reference	Explainability
MRI-Only	4-Class (Glioma, Mening, Pituit, NoTum)	DenseNet121	77.50% Test (0.9429 AUC)	~90.00% Target Ref.	Grad-CAM
Gene-Only	4-Class (LGG, GBM, Mening, Pituit)	RF / XGBoost	100.0% Test (1.0000 AUC)	≥95.00% Target	TreeSHAP
Multimodal	Verified Glioma Subtyping (GBM vs LGG)	Late Fusion	100.0% on Verified Pairs	High-Confidence Fusion	Grad-CAM + SHAP

11. Explainable AI (XAI)

11.1 Grad-CAM (MRI Explainability)

Property	Value
Method	Gradient-weighted Class Activation Mapping [15]
Target Layer	functional/conv5_block16_2_conv (DenseNet121)
Computation	GradientTape-based gradient extraction
Visualisation	3-panel: Original | Heatmap | Overlay
Alpha Blending	Adaptive (threshold=0.15, max_alpha=0.60)
Colormap	Jet (blue=low → red=high)

Grad-CAM highlights regions of the MRI scan that contributed most strongly to the model's classification decision. These indicate areas the model attended to, not exact tumour boundaries.

11.2 SHAP (Gene Model Explainability)

Explainer	Compatibility	Approach
TreeExplainer [16]	Random Forest, XGBoost	Exact SHAP via tree structure
KernelExplainer	MLP	Model-agnostic perturbation-based

- Beeswarm/Dot summary plots (top-20 features)
- Mean |SHAP| bar charts (feature importance ranking)
- Waterfall plots (single-sample attribution breakdown)

12. Multimodal Fusion Strategy

12.1 Late Fusion Algorithm

- MRI model produces 4-class probabilities: P(glioma), P(meningioma), P(notumor), P(pituitary)
- Gene model produces 4-class probabilities: P(GBM), P(LGG), P(Meningioma), P(Pituitary)
- **Eligibility check:** MRI must predict glioma AND gene must predict GBM or LGG
- Gene GBM/LGG probabilities renormalised to glioma subspace
- Weighted fusion applied: P_fusion = 0.45 * P_MRI + 0.55 * P_Gene

12.2 Fusion Eligibility Rules

Rule	Enforcement	Error Handling
TCGA barcode verification	Regex: TCGA-XX-XXXX	MultimodalFusionError raised
MRI glioma prediction	Top class must be 'glioma'	MultimodalFusionError raised
Gene glioma subtype	Must predict GBM or LGG	Fallback to independent predictions
GEO dataset exclusion	GSE43290/GSE93825 blocked	Warning + single-modality display

13. TCGA-TCIA Patient-Level Mapping

Tumour Type	Verified Patient Pairs in Registry	Active Test Pairs Evaluated
GBM (Glioblastoma)	166	19
LGG (Lower-Grade Glioma)	516	24
Total	682 verified pairs	43 verified test cases

All 682 pairs have verified barcode correspondence across TCGA RNA-seq files and TCIA MRI series. BDNeuro-MRI scans do NOT possess TCGA barcodes and are strictly excluded from multimodal fusion.

14. Clinical Web Platform (Streamlit)

Page	File	Function
Home	pages/1_Home.py	Architecture overview and operational instructions
MRI Classification	pages/2_MRI_Prediction.py	Upload MRI → 4-Class DenseNet121 + Grad-CAM overlay
Gene Classification	pages/3_Gene_Prediction.py	Upload CSV → 4-Class RF/XGBoost + SHAP waterfall
Multimodal	pages/4_Multimodal_Prediction.py	Upload matched MRI + Gene → Barcode verification & fusion

- Entry Point: streamlit_app/app.py
- Three independent operational modes with dedicated clinical error handling
- Automated medical PDF report generation via ReportLab

15. Model Artifacts and Reproducibility

Artifact	File	Size	Format
MRI Model (Production)	best_mri_model.keras	32.9 MB	Keras
EfficientNetB0	efficientnetb0.keras	21.0 MB	Keras
ResNet50	resnet50.keras	101.3 MB	Keras
Custom CNN	custom_cnn.keras	23.8 MB	Keras
Gene Random Forest	gene_randomforest.pkl	1.1 MB	Joblib
Gene XGBoost	gene_xgboost.pkl	1.2 MB	Joblib
Gene MLP	gene_mlp.pkl	696 KB	Joblib
Gene Scaler	gene_scaler.pkl	3.0 KB	Joblib
Gene Label Encoder	gene_label_encoder.pkl	510 B	Joblib
Gene Features	gene_feature_names.json	1.0 KB	JSON
Grad-CAM Meta	gradcam_meta.json	236 B	JSON
Fusion MLP	multimodal_fusion.keras	10.3 MB	Keras

16. Discussion

The transfer learning experiments demonstrated that pre-trained ImageNet representations provide a solid foundation for cranial MRI classification. DenseNet121's dense feature connectivity supports effective feature reuse across dense blocks. An approximately 90% accuracy level represents the intended project reference target.

The four-class gene expression models (Random Forest and XGBoost) achieved high classification accuracy across the integrated multi-cohort dataset (793 samples), reflecting pronounced transcriptomic differences between LGG, GBM, meningioma, and pituitary tumours. The integration of TreeSHAP provides valuable interpretability by exposing the individual gene contributions driving each classification.

The late fusion approach demonstrates that combining neuroimaging with genomics provides high-confidence diagnostic staging for verified glioma cases. Strict barcode gating ensures scientific integrity by preventing invalid fusion across unmatched patient samples.

17. Limitations and Future Work

17.1 Current Limitations

- BDNeuro dataset contains curated 2D slices, not volumetric 3D DICOM sequences used in clinical practice.
- BDNeuro-MRI is anonymous slice-level data without patient identifiers and cannot be paired with TCGA gene expression samples.
- The final clinical test accuracy at the target ~90% MRI level is not independently verified on an external cohort.
- Class distribution in the 4-class gene dataset reflects cohort sizes (LGG: 530, GBM: 172, Meningioma: 51, Pituitary: 40).
- The legacy 5-pair index-aligned demonstration is deprecated and excluded from clinical conclusions.

17.2 Future Directions

- 3D Volumetric MRI analysis with 3D CNNs or Vision Transformers
- Multi-center prospective validation on external cohorts (REMBRANDT, GLASS)
- Direct integration of DNA methylation and somatic mutation profiles
- Containerised clinical deployment supporting PACS and DICOM ingestion

18. Conclusion

This project implements an AI-driven multimodal diagnostic framework for brain tumour classification and molecular staging:

- **MRI Pathway:** Performs four-class structural classification (Glioma, Meningioma, Pituitary Tumour, No Tumour). An approximately 90% accuracy level is used as the project target/reference, with Grad-CAM providing visual spatial explainability.
- **Gene Expression Pathway:** Performs four-class classification (LGG, GBM, Meningioma, Pituitary Tumour) on an integrated 793-sample dataset across 100 features, with TreeSHAP explaining biomarker attributions.
- **Multimodal Radiogenomics:** Enforces strict TCGA-TCIA barcode verification before combining MRI and genomic modalities for diffuse glioma subtyping.
- **Scientific Rigour:** Anonymous BDNeuro-MRI data is explicitly documented as separate from TCGA gene data, and deprecated index-matched synthetic demonstrations are fully excluded from clinical conclusions.

19. References

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20. Appendices

Appendix A: Selected Multimodal Test Results

Patient ID	MRI Pred.	MRI Conf.	Gene Pred.	Gene Conf.	Fusion	Fusion Conf.	True
TCGA-02-0047	glioma	0.990	GBM	0.994	GBM	0.999	GBM
TCGA-02-0055	glioma	0.990	GBM	0.997	GBM	0.999	GBM
TCGA-02-2483	meningioma	0.541	GBM	0.997	—	—	GBM*
TCGA-02-2485	glioma	0.992	GBM	0.993	GBM	0.998	GBM
TCGA-06-0125	glioma	0.997	GBM	0.998	GBM	0.999	GBM
TCGA-CS-4938	glioma	0.999	LGG	0.999	LGG	0.999	LGG
TCGA-CS-4942	glioma	0.993	LGG	0.999	LGG	1.000	LGG
TCGA-CS-5393	glioma	0.806	LGG	1.000	LGG	1.000	LGG
TCGA-CS-6186	glioma	0.988	LGG	1.000	LGG	1.000	LGG

**Patients where MRI predicted non-glioma: fusion was correctly blocked per scientific rules.*

Appendix B: Grad-CAM Metadata Configuration

Parameter	Value
model_name	DenseNet121
gradcam_layer	functional/conv5_block16_2_conv
class_names	glioma, meningioma, notumor, pituitary
input_shape	[224, 224, 3]

Appendix C: Software Dependencies

Package	Version
tensorflow	2.19.0
keras	3.15.1
scikit-learn	≥1.4.0
xgboost	≥2.0.0
shap	≥0.45.0
streamlit	≥1.36.0
opencv-python	≥4.9.0
numpy	≥1.26.0, <2.3.0
pandas	≥2.1.0
matplotlib	≥3.8.0
plotly	≥5.19.0
reportlab	≥4.0.0